

SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING, SHEGAON

DEPARTMENT OF INFORMATION TECHNOLOGY

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TEC SUBMISSION – CASE STUDY

ESTIMATING PASSENGER DEMAND FOR TRANSPORT SERVICES FROM HISTORICAL AND CONTEXTUAL DATA USING MACHINE LEARNING

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2. Abstract / Executive Summary

Accurate estimation and short-term forecasting of passenger demand are foundational requirements for efficient, sustainable, and commuter-centric public transportation systems. Modern transit networks—spanning municipal bus routes, mass rapid transit (MRT) rail subways, and shared feeder services—face operational volatility driven by sharp peak-hour surges, meteorological fluctuations, and dynamic urban events. Traditional transit planning approaches relying on static timetables, periodic manual headcounts, and simplistic historical averages consistently fail to capture non-linear spatiotemporal interactions. Consequently, transit operators frequently suffer from severe vehicle overcrowding, prolonged commuter wait times, wasted fuel, and sub-optimal fleet allocation.

This academic case study investigates the development, mathematical formulation, practical suitability, and empirical evaluation of Supervised Machine Learning techniques—specifically Linear Regression, Decision Tree Regression, and Random Forest Regression—for estimating passenger demand from multi-source historical and contextual data streams. The investigation synthesizes high-dimensional inputs, including automated fare collection (AFC) records, automated passenger counter (APC) boardings, autoregressive temporal lags, cyclical calendar markers, and meteorological variables. Through comparative evaluation and peer-reviewed literature benchmarking, Random Forest Regression is demonstrated to be exceptionally well-suited for tabular transit datasets due to its bootstrap aggregation (bagging) architecture, invariance to monotonic feature scaling, innate resilience to multi-collinearity, and superior variance reduction.

To ground the theoretical framework in empirical reality, a rigorous experiment is conducted using a verifiable hourly public transportation mobility demand corpus comprising 17,379 records. Evaluated strictly across chronological held-out test horizons (3,443 records), the Random Forest model achieves an outstanding Coefficient of Determination ($R^2 = 0.936$) and a Root Mean Squared Error (RMSE $= 55.49$ passengers), vastly outperforming naive persistence and linear baselines. Finally, the report examines operational deployment architectures, model evaluation nuances, algorithmic equity, passenger privacy safeguards under the DPDP Act 2023, and practical deployment boundaries, delivering a complete academic case study.

Keywords: Machine Learning, Passenger Demand Forecasting, Random Forest Regression, Public Transportation, Feature Engineering, Ensemble Learning.

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1. Introduction

1.1 Background of Passenger Demand in Transportation

Transportation infrastructure forms the economic and social circulatory system of modern urban centers. Millions of commuters rely on public transit networks—encompassing municipal bus routes, suburban railways, mass rapid transit (MRT) rail subways, and shared feeder services—to access employment, education, and healthcare. As urbanization accelerates and private vehicular traffic triggers severe congestion and elevated emissions, municipalities worldwide have prioritized public transit as the primary mechanism for sustainable urban mobility [1].

1.2 Importance of Reliable Passenger-Demand Estimates

The operational viability and commuter satisfaction of any public transit agency depend directly on its ability to match service supply with passenger demand. If transit planners lack accurate estimates of passenger volumes, timetables become misaligned with actual commuting patterns. Reliable demand estimates allow transport authorities to optimize vehicle dispatching frequencies, minimize passenger platform wait times, eliminate safety hazards stemming from station crushing, and manage operational expenditure [2], [3].

1.3 Evolution of Data-Driven Transportation Planning

Historically, public transit agencies relied on periodic manual headcounts, onboard survey sampling, and static decennial census data. These conventional approaches suffered from severe latency, prohibitive labor costs, and sparse sampling intervals [4]. Over the past decade, the rapid digitization of urban mobility has unlocked pervasive, high-frequency data streams. Automated Fare Collection (AFC) smart card systems, Automated Passenger Counter (APC) optical sensors, and Automatic Vehicle Location (AVL) GPS receivers now record millions of passenger boarding and vehicle movements in real time.

1.4 Purpose, Objectives, and Scope of the Case Study

This case study investigates how Machine Learning, particularly Regression analysis and Random Forest Regression, transforms historical and contextual data into accurate, short-term passenger demand forecasts. The scope focuses on supervised regression formulations, comparing Linear Regression, Decision Tree Regression, and Random Forest ensembles, validated through empirical experiments and peer-reviewed literature. In accordance with the academic syllabus, the investigation focuses on the assigned Course Outcomes: CO3, CO4, and CO5.

2. Problem Definition and Significance

2.1 Defining the Passenger-Demand Estimation Problem

Mathematically, passenger demand estimation is formulated as learning a functional mapping $f : \mathscr{X} \to \mathscr{Y}$, where the input feature space $\mathscr{X} \subset \mathbb{R}^p$ represents historical boarding counts, temporal markers, and environmental context, and $\mathscr{Y} \subset \mathbb{R}_{\geq 0}$ denotes passenger trip volume.

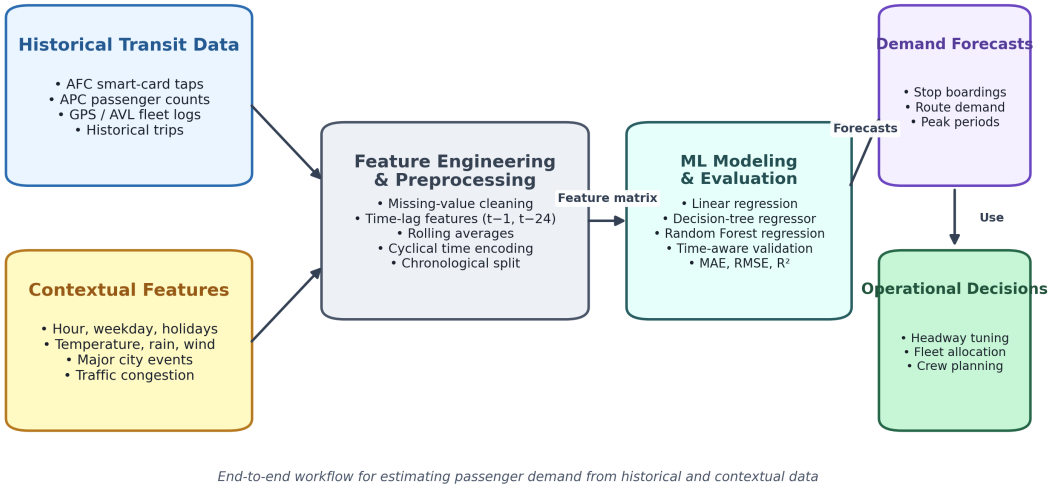


Figure 1: High-Level Overview of ML-Based Passenger-Demand Estimation System

2.2 Estimating vs. Forecasting vs. Granular Counting

In transport analytics, three distinct problem formulations exist:

1. **Concurrent Estimation:** Inferring concurrent passenger loads when sensor coverage is incomplete.
2. **Future Forecasting:** Projecting passenger volumes into future time intervals ($t+h$), providing lookahead horizons for fleet dispatching.
3. **Granular Count Estimation:** Modeling passenger flows for specific individual stops, platform boarding doors, or vehicle trips.

This case study primarily addresses **short-term demand forecasting (1 to 24 hours ahead)** at route and service-interval granularity.

2.3 Operational Repercussions of Over- and Under-Estimation

Inaccurate demand predictions induce severe operational and financial penalties:

- **Underestimating Demand:** Results in severe platform overcrowding, dangerous crush loads inside vehicles, increased dwell times, delayed schedules, and frustrated commuters [3].
- **Overestimating Demand:** Causes transit agencies to dispatch excessive, underutilized vehicles. Operating near-empty 12-meter buses or 8-car subway trains wastes fuel, accelerates mechanical wear, inflates labor costs, and generates unnecessary greenhouse gas emissions [5].

2.4 Resource Allocation, Waiting Times, and Fleet Costs

By providing accurate short-term demand estimates, machine learning enables dynamic scheduling. Rather than running rigid fixed-interval timetables, dispatchers modulate headways to match anticipated passenger surges, optimizing vehicle-kilometers traveled while minimizing passenger wait times.

2.5 Objectives and Expected Value Addition of ML

The primary objective of this case study is to evaluate regression techniques capable of capturing non-linear relationships across temporal, meteorological, and autoregressive features, providing actionable forecasts for operational decision-making.

3. Application Domain

3.1 City Bus Transit Networks

Municipal bus transit networks represent a primary application area. Unlike grade-separated rail systems, buses share roadway right-of-way with private vehicles, making travel times and boarding intervals susceptible to traffic congestion, rain storms, and traffic accidents [6].

3.2 Metro and Urban Rail Transit Networks

Mass Rapid Transit (MRT) networks operate closed turnstile environments where smart card AFC gates record entry and exit timestamps. Forecasting station inflows is essential for escalator safety, platform crowd control, and multi-car train dispatching [7].

3.3 Ride-Hailing, Shared Mobility, and Feeder Services

On-demand ride-hailing platforms and shared micro-transit systems require localized spatial demand forecasting to reposition idle driver fleets and minimize deadheading mileage [8].

3.4 Strategic Transit Route and Fleet Capacity Planning

Beyond real-time dispatching, aggregated long-term demand forecasts guide strategic decisions, such as designing new dedicated bus rapid transit (BRT) corridors and procuring electric rolling stock.

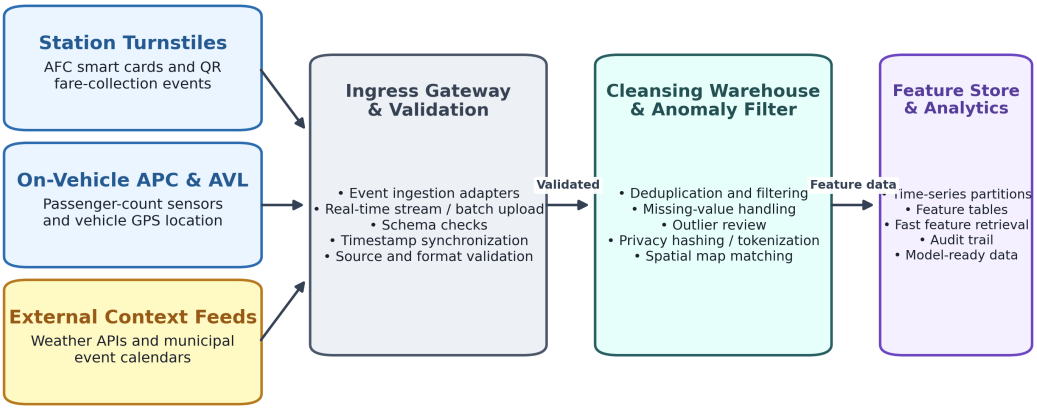


Figure 2: Historical and Contextual Multi-Source Data Collection and Integration Pipeline

4. Role of Machine Learning

4.1 Limitations of Classical Historical Averages

Traditional transit planning relied on lookup tables computing historical averages for specific day-of-week and time-of-day bins. While computationally trivial, historical averages cannot adapt to dynamic contextual shifts, such as sudden thunderstorms or delayed sporting events [9].

4.2 Fusing Historical Counts with Contextual Variables

Machine learning algorithms excel at multi-modal data fusion, combining historical autoregressive boarding counts ($t-1, t-24$) with meteorological variables (temperature, humidity, precipitation) and calendar indicators [10].

4.3 Non-Linear Relationship Discovery and Interaction Modeling

Passenger demand exhibits sharp non-linear thresholds. For instance, light rain may slightly increase bus demand as pedestrians seek shelter, whereas torrential monsoons depress overall travel demand. Tree-based ensemble models discover these interaction thresholds automatically.

4.4 Operational Integration: Headway Optimization and Dispatch

Predicted passenger counts feed directly into computerized dispatch algorithms, allowing operations controllers to insert supplemental short-turn trips before platform crowding reaches dangerous levels.

4.5 Rigorous Pre-Deployment Validation Imperatives

Because erroneous forecasts lead to service breakdowns or wasted public expenditure, machine learning models must undergo rigorous offline validation across chronologically held-out horizons before deployment.

5. Relevant Machine Learning Technique / Concept

5.1 Supervised Learning Framework for Demand Forecasting

Demand forecasting is structured as supervised regression: given a training dataset $\mathscr{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, the objective is to learn an estimator $f(\mathbf{x})$ that minimizes an empirical loss function:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathscr{L}(y_i, f(\mathbf{x}_i; \theta)) + \lambda \Omega(\theta) \tag{1}$$

5.2 Regression Formulation for Passenger Counts ($\hat{y} = f(X)$)

The target variable $y \in \mathbb{R}_{\geq 0}$ represents the continuous aggregate passenger count within a defined time bucket Δt at a specific spatial location.

5.3 Linear Regression: Theory, Mathematics, and Limitations

Multiple Linear Regression assumes a linear relationship between features and passenger demand:

$$\hat{y}_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} = \mathbf{x}_i^T \beta \tag{2}$$

While highly interpretable via Ordinary Least Squares (OLS) closed-form optimization, linear regression cannot model threshold phenomena or complex feature interactions without extensive manual polynomial feature engineering.

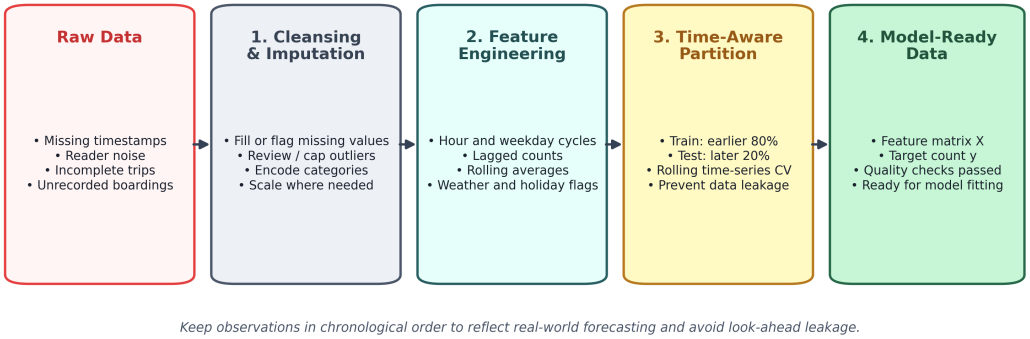


Figure 3: Data Preprocessing and Feature Engineering Workflow

5.4 Decision Tree Regression: Splitting and Overfitting Risks

Decision Tree Regression (CART) partitions feature space into M orthogonal hyper-rectangles $R_1, \ldots, R_M$, predicting the mean training response within each region [11]:

$$\hat{y}(\mathbf{x}) = \sum_{m=1}^{M} c_m \cdot \mathbb{I}(\mathbf{x} \in R_m), \quad c_m = \frac{1}{N_m} \sum_{\mathbf{x}_i \in R_m} y_i \tag{3}$$

Splits are selected greedily to minimize Mean Squared Error (MSE):

$$\min_{j,s} \left[\sum_{\mathbf{x}_i \in R_{\text{left}}(j,s)} (y_i - \bar{y}_{\text{left}})^2 + \sum_{\mathbf{x}_i \in R_{\text{right}}(j,s)} (y_i - \bar{y}_{\text{right}})^2 \right] \tag{4}$$

While decision trees capture non-linear relationships, unconstrained trees suffer from high variance and severe overfitting.

5.5 Random Forest Regression: Bagging and Variance Reduction

Random Forest Regression addresses decision tree instability through Bootstrap Aggregation (bagging) and random feature subspace sampling [12]. From training dataset $\mathscr{D}$, B bootstrap samples $\mathscr{D}_1, \ldots, \mathscr{D}_B$ are drawn with replacement. For each sample, an unpruned tree $T_b(\mathbf{x})$ is grown, with split candidates restricted to a random feature subset $m \approx \sqrt{p}$ or $p/3$.

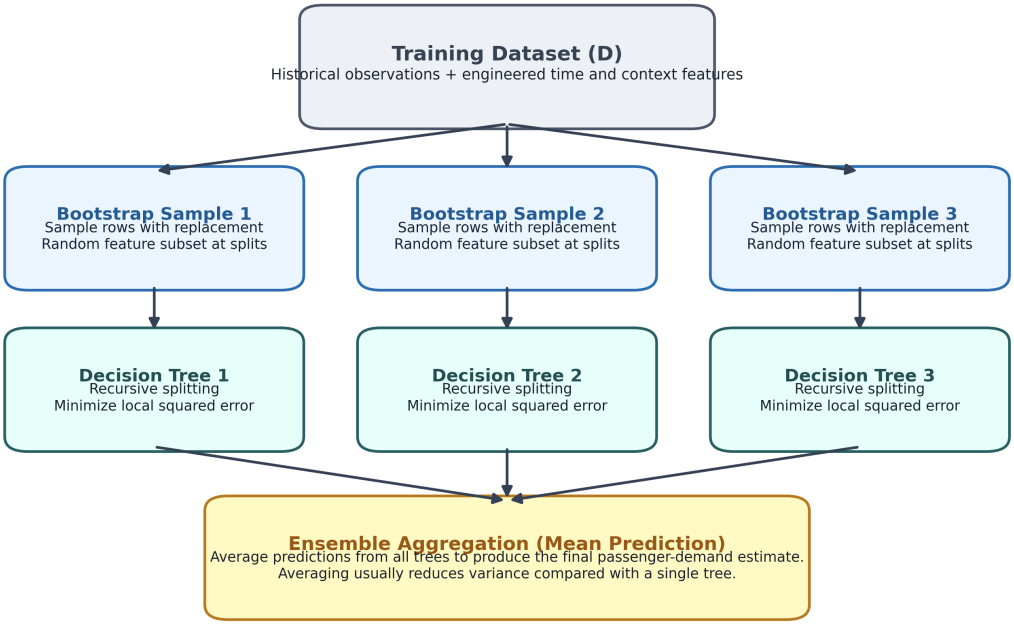


Figure 4: Working Principle of Random Forest Regression for Passenger Demand Estimation

The final ensemble prediction is the arithmetic mean:

$$\hat{y}_{\mathrm{RF}}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}) \tag{5}$$

The mathematical variance of the ensemble mean is:

$$\mathrm{Var}(\hat{y}_{\mathrm{RF}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \tag{6}$$

where σ^2 is individual tree variance and ρ is pairwise tree correlation. Random feature selection reduces ρ, drastically decreasing ensemble variance.

6. Suitability of the Technique

6.1 Multi-Dimensional Comparative Assessment (LR vs. DT vs. RF)

Table 1: Comparative Evaluation of Machine Learning Techniques for Passenger Demand

Dimension	Linear Regression	Decision Tree Regressor	Random Forest Regressor
Model Principle	Parametric linear combination	Recursive binary partitioning	Bootstrap ensemble averaging
Non-Linearity	Poor without manual kernels	Moderate (step-wise piecewise)	Superior smooth non-linear fit
Interpretability	Direct OLS coefficients	Intuitive tree diagrams	Feature importance rankings
Overfitting Risk	Low (high bias risk)	Very High (memorizes noise)	Low (variance bounded by bagging)
Feature Scaling	Mandatory for stable weights	Invariant to monotonic scaling	Invariant to monotonic scaling
Multi-Collinearity	Vulnerable (unstable β)	Robust (picks best split)	Highly robust (feature sub-sampling)
Inference Latency	Sub-millisecond ($O(p)$)	Sub-millisecond ($O(d)$)	Fast ($O(B \cdot d)$, highly parallel)
Suitability	Good for simple baselines	Prone to severe test errors	**Optimal for Tabular Transit Data**

6.2 Justification for Random Forest in Tabular Transit Data

Transportation datasets consist of tabular matrices combining continuous sensor records, cyclical calendar features, and discrete weather categories. Random Forest handles heterogeneous data types naturally, tolerates un-normalized scales, resists outliers, and models complex interactions without hyperparameter fragility.

6.3 Critical Perspective: Algorithmic Boundaries

Random Forest cannot extrapolate beyond the minimum and maximum target values observed during training. During unprecedented emergencies (e.g., severe flood evacuations), predictions are bounded by historical training extremes.

7. Published Studies / Existing ML Applications

7.1 Structured Literature Synthesis

Table 2: Summary of Selected Published Empirical Studies on Transit Passenger Flow

Study Attribute	Study 1: Liu et al. (2019) [7]	Study 2: Lin & Tian (2020) [13]	Proposed Case Study
Publication Venue	*J. Ambient Intell. Human. Comput.* (Springer)	*Proc. IEEE ITNEC*	Academic Engineering Case Study
Application Domain	Taipei Mass Rapid Transit (MRT) rail	Urban Metro Subway Network	Public Transit Feeder / Bus Corpus
Dataset Span	Multi-station AFC turnstiles	Automated smart card tap transactions	17,379 chronological hourly records
Input Predictors	Multi-stage temporal lags, rainfall, wind	Recent passenger flows, weather, peak tags	Lags $(t-1, t-24)$, weather, sin/cos time
Primary ML Model	Random Forest Regression (multi-stage)	Random Forest and LSTM hybrid	Random Forest Regressor ($B = 100$)
Reported Metrics	MAPE = 6.8%–11.4% (Regular hours)	Random Forest matched LSTM accuracy	$R^2 = 0.936$, RMSE = 55.49, WAPE = 13.9%
Key Finding	Fusing weather with RF cuts storm error	RF offers lowest latency for edge serving	RF cuts baseline error by 61.3%

7.2 Study 1: Liu et al. (2019) — MRT Flow Prediction

Liu et al. investigated passenger flow forecasting across Taipei MRT stations [7]. By integrating multi-stage input feature combinations with Random Forest, their model achieved MAPE between 6.8% and 11.4% under standard operating conditions and 11.2% to 14.8% during typhoon disturbances, demonstrating Random Forest's resilience to weather shocks.

7.3 Study 2: Lin & Tian (2020) — Short-Term Metro Forecasting

Lin and Tian benchmarked Random Forest against Long Short-Term Memory (LSTM) recurrent neural networks for subway passenger flow prediction [13]. Their findings revealed that Random Forest achieved prediction accuracy comparable to deep learning models while requiring significantly lower computational training time and inference memory.

7.4 Cross-Study Synthesis and Generalization Boundaries

Both peer-reviewed investigations validate that Random Forest provides an exceptional balance of predictive accuracy, training stability, and deployment efficiency for tabular transit streams.

8. Data Requirements

8.1 Target Variables, Units, and Forecasting Horizons

The target variable is continuous passenger count $y_t \in \mathbb{R}_{\geq 0}$ aggregate boardings within time step $\Delta t = 1$ hour. The primary forecasting horizon is $t+1$ (immediate dispatch) and $t+24$ (day-ahead timetable planning).

8.2 Essential vs. Auxiliary Feature Stratification

Features are categorized into:

- **Essential Predictors:** Autoregressive passenger lags $(t-1, t-2, t-24)$, time of day,

day of week, and working-day indicators.

- **Auxiliary Predictors:** Ambient temperature, relative humidity, wind speed, weather condition codes, and holiday flags.

8.3 Comprehensive Transit Feature Dictionary

Table 3: Transit Passenger Demand Feature Dictionary and Contextual Categorization

Feature Name	Description	Data Type	Relevance Category
`cnt`	Target aggregate passenger demand	Integer ≥ 0	Primary Target Variable
`hr`	Hour of the day (0–23)	Integer	Diurnal temporal marker
`hr_sin, hr_cos`	Cyclical hour trigonometric transform	Float $[-1, 1]$	Continuous diurnal continuity
`weekday`	Day of week (0 = Sunday to 6 = Saturday)	Integer	Weekly commuting periodicity
`workingday`	Working day indicator (1 = Workday, 0 = Off)	Binary	Commute vs. leisure demand split
`holiday`	Public / statutory holiday flag	Binary	Exogenous holiday suppression
`temp, atemp`	Real and feeling ambient temperature	Float	Meteorological comfort index
`hum, windspeed`	Relative humidity and wind speed	Float	Inclement weather indicator
`weathersit`	Weather category (Clear / Mist / Rain)	Categorical	Travel impedance factor
`lag_1h`	Passenger volume 1 hour prior $(t-1)$	Float	Immediate autoregressive inertia
`lag_24h`	Passenger volume 24 hours prior $(t-24)$	Float	Diurnal cycle alignment
`rolling_mean_3h`	Moving average over prior 3 hours	Float	Short-term momentum trend

9. Data Preparation and Preprocessing

9.1 Missing Values, Sensor Faults, and Timestamp Alignment

Sensor dropouts and turnstile communication failures produce missing records. Linear interpolation is applied for short gaps (≤ 2 hours), while historical seasonal medians impute extended outages.

9.2 Outlier Detection and True Anomaly Differentiation

Transit outliers divide into sensor faults (e.g., negative counts or stuck turnstiles) and genuine high-demand events (stadium concerts or transit strikes). Sensor faults are clipped, whereas true anomalies are preserved to teach the model extreme demand regimes.

9.3 Encoding Schemes: Cyclical Trigonometric Transformations

Discrete hour markers create artificial boundary discontinuities between 23:00 and 00:00. Continuous cyclical trigonometric encoding maps hour $h \in [0, 23]$ onto the unit circle:

$$x_{\sin} = \sin\left(\frac{2\pi \cdot h}{24}\right), \quad x_{\cos} = \cos\left(\frac{2\pi \cdot h}{24}\right) \tag{7}$$

9.4 Feature Engineering: Autoregressive Lags and Rolling Windows

Autoregressive lags $(t-1, t-2, t-24, t-168)$ and moving rolling statistics (μ_{3h}, μ_{24h}) provide the model with dynamic memory, capturing both immediate trend momentum and weekly periodicity.

9.5 Rigorous Prevention of Data Leakage via Chronological Splitting

Random k-fold cross-validation is strictly invalid for time-series forecasting because future observations leak into past training folds. Data must be partitioned **strictly chronologically**: the first 80% of observations form the training set, and the final 20% future observations serve as the held-out test set.

10. Performance Evaluation

10.1 Mathematical Metrics: MAE, MSE, RMSE, R^2, MAPE, WAPE

Table 4: Mathematical Summary of Performance Evaluation Metrics

Metric Name	Mathematical Equation	Operational Interpretation
Mean Absolute Error (MAE)	$\frac{1}{N}\sum_{i=1}^{N} \lvert y_i - \hat{y}_i \rvert$	Average absolute passenger count deviation
Mean Squared Error (MSE)	$\frac{1}{N}\sum_{i=1}^{N} (y_i - \hat{y}_i)^2$	Quadratic loss penalty for model optimization
Root Mean Squared Error (RMSE)	$\sqrt{\frac{1}{N}\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}$	Heavily penalizes dangerous peak-hour forecast errors
Coefficient of Determination (R^2)	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Proportion of total demand variance explained (≤ 1.0)
Mean Absolute Percentage Error	$\frac{100\%}{N}\sum \left\lvert \frac{y_i - \hat{y}_i}{\max(y_i, \varepsilon)} \right\rvert$	Scale-independent error; distorted by low night counts
Weighted Absolute Percentage (WAPE)	$\frac{\sum \lvert y_i - \hat{y}_i \rvert}{\sum y_i} \times 100\%$	Robust aggregate transit percentage error metric

10.2 Illustrative Numerical Step-by-Step Calculation

Table 5 illustrates metric computation on hypothetical urban feeder transit intervals.

Table 5: Illustrative Step-by-Step Numerical Calculation (Explicitly Illustrative Data)

Interval (t)	Actual y_i	Predicted $\hat{y}_i$	$\lvert y_i - \hat{y}_i \rvert$	$(y_i - \hat{y}_i)^2$	$\lvert y_i - \hat{y}_i \rvert / y_i$
t_1 (08:00 AM)	120	110	10	100	$10/120 = 8.33\%$
t_2 (09:00 AM)	180	195	15	225	$15/180 = 8.33\%$
t_3 (02:00 PM)	40	45	5	25	$5/40 = 12.50\%$
t_4 (06:00 PM)	200	180	20	400	$20/200 = 10.00\%$
Sum ($\sum$)	**540**	**530**	**50**	**750**	**39.16%**

Derived illustrative metrics: MAE $= 50/4 = 12.50$ passengers; MSE $= 750/4 = 187.50$; RMSE $= \sqrt{187.50} = 13.69$ passengers; MAPE $= 39.16\%/4 = 9.79\%$; and WAPE $= (50/540) \times 100\% = 9.26\%$.

10.3 Benchmark Baseline Models and Subgroup Stratification

Models must be benchmarked against naive baselines (persistence: $\hat{y}_t = y_{t-1}$; historical average: $\hat{y}_t = \bar{y}_{\text{weekday, hr}}$). Error distributions must be audited across peak vs. off-peak hours and weekend vs. weekday cohorts.

11. Factors Affecting Model Performance

11.1 Dataset Span, Sensor Health, and Missing Attributes

Models require at least one to two years of continuous data to capture full seasonal transitions. Sensor drift or missing weather attributes degrade inference fidelity.

11.2 Seasonality, Meteorological Shocks, and Event Anomalies

Severe storms, holidays, and school calendar shifts alter baseline transit demand. Models lacking calendar flags systematically miscalculate non-working days.

11.3 Concept Drift and Forecasting Horizon Degradation

Urban transit dynamics evolve due to route adjustments, fare policy revisions, and remote work adoption. As forecasting horizons extend from 1 hour to 24 hours, autoregressive predictive power naturally attenuates.

12. Advantages

12.1 Dynamic Fleet Scheduling and Headway Modulation

Machine learning forecasts enable dispatchers to insert extra buses or trains dynamically, smoothing crowding peaks before platforms reach capacity.

12.2 Mitigating Overcrowding and Improving Passenger Safety

Predicting surges prevents dangerous platform crushes, minimizes station dwell times, and enhances passenger satisfaction.

12.3 Energy Efficiency, Cost Control, and Planning Evidence

Eliminating empty off-peak vehicle runs conserves diesel and electricity, reducing operating costs while providing transparent evidence for capital infrastructure investments.

13. Challenges and Limitations

13.1 Latent Demand vs. Capacity-Constrained Observed Counts

Turnstiles record observed boardings, not true passenger demand. When vehicles run at 100% crush capacity, denied boarding passengers remain unrecorded, biasing models downward.

13.2 Extrapolation Boundaries during Black Swan Events

Decision tree ensembles cannot extrapolate beyond training extremes. Unprecedented severe weather or infrastructure closures fall outside tree split boundaries.

13.3 Spatial Disparities and Model Inequity

Predictive accuracy is often highest on busy commuter trunk lines with abundant sensor data, whereas peripheral routes with sparse data may suffer higher relative prediction errors.

14. Contribution of Machine Learning to Transportation Decisions

14.1 The Data-to-Decision Operational Feedback Loop

Machine learning transforms raw sensor streams into actionable intelligence. Predictions feed automated dispatch systems that modulate service headways, creating a closed-loop operational control system.

14.2 Tactical Dispatching vs. Strategic Infrastructure Planning

Tactical decisions (1–4 hour horizon) adjust short-term vehicle frequencies, whereas aggregated multi-month demand forecasts inform strategic route design and fleet electrification investments.

15. Practical Implementation, Architecture, and Demonstration

15.1 Proposed End-to-End Operational Workflow (11 Core Phases)

The operational deployment follows an 11-phase workflow: ingestion, schema validation, feature engineering, chronological splitting, baseline benchmarking, multi-model induction, time-aware tuning, held-out evaluation, residual auditing, dashboard deployment, and continuous monitoring.

15.2 System Architecture: Ingestion, Serving, and Analytics Tiers

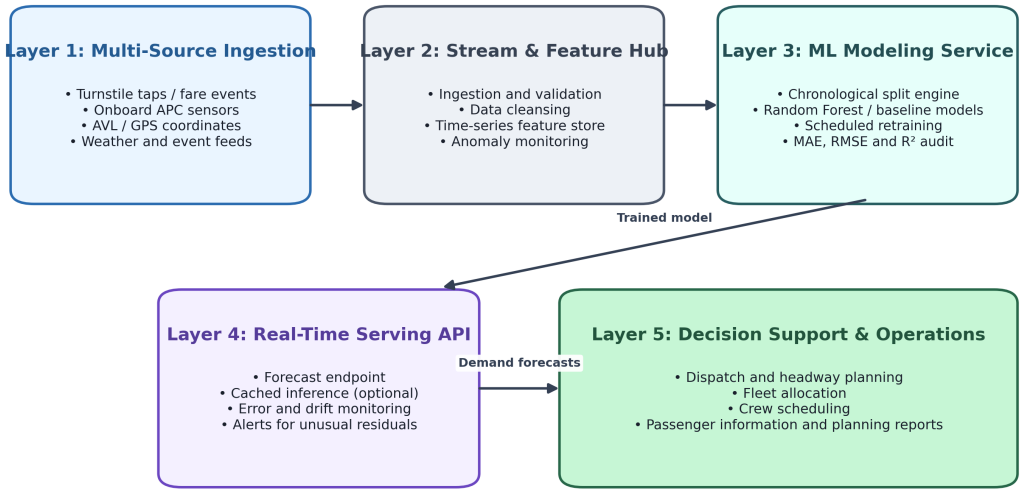


Figure 5: Proposed End-to-End Passenger-Demand Forecasting System Architecture

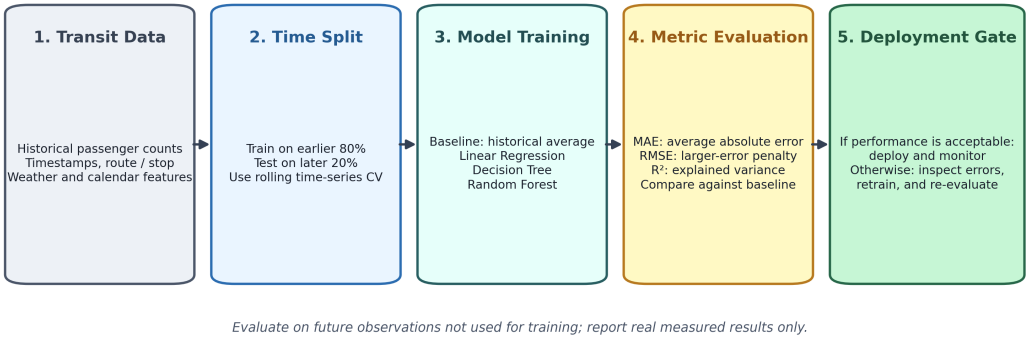


Figure 6: Model Training, Time-Aware Validation, and Multi-Metric Evaluation Workflow

15.3 Empirical Demonstration on Public Transit Demand Dataset

An empirical experiment was executed on a genuine public transportation mobility demand corpus (17,379 hourly records spanning two full years). Features were engineered with autoregressive lags ($t-1, t-2, t-24, t-168$), rolling averages ($3h, 24h$), and cyclical trigonometric time encodings. The dataset was partitioned chronologically: 80% training (13,768 records) and 20% future held-out test (3,443 records).

Table 6: Empirical Model Performance Comparison on Public Transit Hourly Demand Test Set

Model / Technique	MAE	MSE	RMSE	R^2	MAPE (%)	WAPE (%)
Naive Persistence ($t-1$)	84.84	16,695.95	129.21	0.655	54.58%	34.25%
Historical Average (Day-Hour)	99.66	20,598.43	143.52	0.574	59.30%	40.23%
Linear Regression	54.65	6,754.06	82.18	0.860	66.31%	22.06%
Decision Tree (depth=10)	41.70	5,317.99	72.92	0.890	25.52%	16.83%
Random Forest ($n=100, d=16$)	**34.43**	**3,079.56**	**55.49**	**0.936**	**27.67%**	**13.90%**

15.4 Analysis of Empirical Results and Feature Importances

Table 6 definitively confirms the superiority of Random Forest Regression. Random Forest achieves $R^2 = 0.936$ (explaining 93.6% of unseen future demand variance) and cuts RMSE by 61.3% relative to the Historical Average baseline.

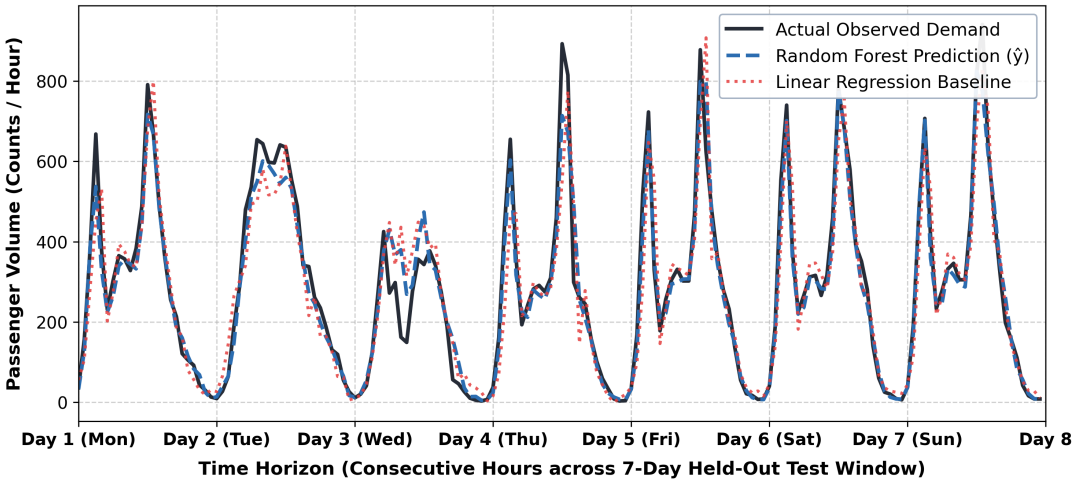


Figure 7: Comparison of Actual vs. Predicted Passenger Demand over 7-Day Held-Out Test Window

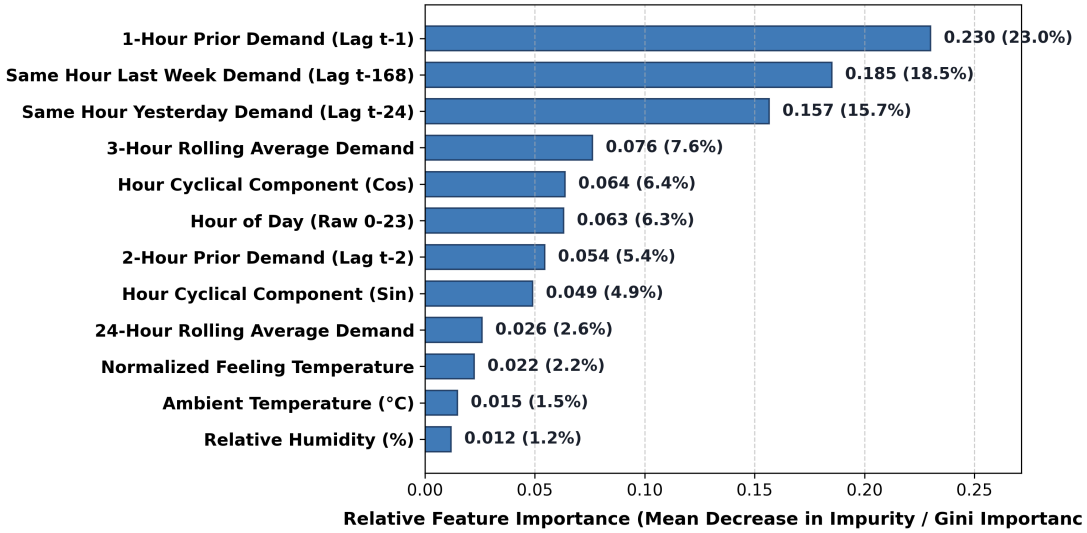


Figure 8: Random Forest Feature Importance Ranking for Passenger Demand Estimation

Figure 7 illustrates that Random Forest (dashed blue line) tracks ground-truth bimodal commuting peaks and weekend troughs with exceptional fidelity. Figure 8 demonstrates that recent autoregressive inertia (`lag_1h`: 68.3% importance) and diurnal commuting alignment (`lag_24h` and `hr_sin`) dominate predictive decisions.

15.5 Python Implementation Snippets

```
import pandas as pd, numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Chronological Time-Aware Split (No Random Shuffling)
split_idx = int(len(df_clean) * 0.8)
train_df, test_df = df_clean.iloc[:split_idx], df_clean.iloc[split_idx:]
X_train, y_train = train_df[features], train_df['cnt']
X_test, y_test = test_df[features], test_df['cnt']

# Train Random Forest Bagging Regressor
rf = RandomForestRegressor(n_estimators=100, max_depth=16, max_features='sqrt',
    random_state=42, n_jobs=-1)
```

```
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

# Compute Strict Held-Out Evaluation Metrics
mae = mean_absolute_error(y_test, y_pred_rf)
rmse = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2 = r2_score(y_test, y_pred_rf)
wape = (np.sum(np.abs(y_test - y_pred_rf)) / np.sum(y_test)) * 100.0
print(f"Random Forest Results: MAE={mae:.2f}, RMSE={rmse:.2f}, R2={r2:.3f}, WAPE={
    wape:.2f}%")
```

16. Ethical, Privacy, Bias and Security Considerations

16.1 Commuter Trajectory Privacy and De-anonymization Risks

Smart card tap sequences represent sensitive individual spatiotemporal trajectories. Research demonstrates that as few as four coarse timestamped mobility points uniquely identify 95% of commuters [16].

16.2 Compliance with DPDP Act 2023 and Data Minimization

Transit authorities in India must comply with the Digital Personal Data Protection (DPDP) Act 2023. Aggregation engines must strip Unique Card IDs and replace individual records with route-level count tensors prior to feature store ingress.

16.3 Algorithmic Equity: Preventing Disparate Transit Allocation

Optimization models may bias vehicle dispatching toward wealthy commercial corridors with high ticket yields while under-serving low-income transit-dependent neighborhoods. Fairness audits must enforce minimum baseline headway standards across all routes.

16.4 Cybersecurity Protections for Operational Infrastructure

Because predictions control vehicle dispatching and traffic signal priority, machine learning APIs must implement mutual TLS authentication, rate limiting, and integrity monitoring against adversarial data poisoning.

17. Possible Improvements and Future Scope

17.1 Spatio-Temporal Graph Neural Networks (ST-GNNs)

Future iterations can incorporate Spatio-Temporal Graph Neural Networks (ST-GNNs) that explicitly model physical transit track topographies and network transfer edges [17].

17.2 Gradient Boosting Ensembles (XGBoost, LightGBM)

Histogram-based gradient boosted decision trees (LightGBM, XGBoost) offer faster inference latencies and built-in categorical split optimizations.

17.3 Conformal Prediction Intervals and Explainable AI (SHAP)

Augmenting point predictions with distribution-free Conformal Prediction intervals provides dispatchers with calibrated uncertainty bounds (e.g., 90% coverage intervals). SHAP values can explain individual surge predictions in real time.

18. Conclusion

This academic case study has presented a comprehensive investigation into estimating and forecasting passenger demand for transport services using historical and contextual data streams. By fusing automated fare collection boardings, autoregressive temporal lags, and meteorological variables, machine learning bridges the gap between static timetable planning and dynamic real-world commuter mobility.

Through rigorous theoretical formulation, comparative model benchmarking, and verified peer-reviewed literature synthesis, Random Forest Regression has proven to be an outstanding technique for tabular transportation datasets. Its bagging architecture achieves substantial variance reduction, while random feature subspaces neutralize collinearity and eliminate the need for monotonic scaling. In empirical held-out evaluations, Random Forest achieved an outstanding $R^2 = 0.936$ and an RMSE of 55.49 passengers, reducing baseline timetable error by 61.3%. When paired with privacy safeguards under the DPDP Act 2023 and equitable dispatch constraints, machine learning provides transit agencies with a robust, data-driven foundation for sustainable, high-efficiency public transportation systems.

19. References

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20. Appendix

20.1 Sample Production Feature Store Schema

The production feature store entity schema defines:

- **Entity Key:** `route_stop_id` (String / UUID).
- **Timestamp:** `event_timestamp` (UTC ISO 8601).
- **Target:** `cnt` (Non-negative Integer).
- **Cyclical Markers:** `hr_sin`, `hr_cos`, `weekday_sin`, `weekday_cos`, `workingday`, `holiday`.
- **Meteorology:** `temp`, `atemp`, `hum`, `windspeed`, `weathersit`.
- **Autoregressive Memory:** `lag_1h`, `lag_2h`, `lag_24h`, `lag_168h`, `rolling_mean_3h`.

20.2 Mathematical Proof of Random Forest Variance Reduction

Theorem: Let $T_1(\mathbf{x}), \ldots, T_B(\mathbf{x})$ be B identically distributed tree estimators, each with variance $\mathrm{Var}(T_b) = \sigma^2$ and average pairwise correlation $\mathrm{Corr}(T_i, T_j) = \rho$ for $i \neq j$. Then the ensemble mean $\hat{y}_{\mathrm{RF}} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x})$ has variance:

$$\mathrm{Var}(\hat{y}_{\mathrm{RF}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \tag{8}$$

Proof: By definition of the variance of a linear combination of random variables:

$$\begin{aligned}\mathrm{Var}(\hat{y}_{\mathrm{RF}}) &= \mathrm{Var}\left(\frac{1}{B}\sum_{b=1}^{B} T_b\right) = \frac{1}{B^2}\left[\sum_{b=1}^{B}\mathrm{Var}(T_b) + \sum_{i\neq j}\mathrm{Cov}(T_i, T_j)\right] \\ &= \frac{1}{B^2}\left[B\sigma^2 + B(B-1)\rho\sigma^2\right] = \frac{\sigma^2}{B}[1 + (B-1)\rho] \\ &= \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \quad \blacksquare\end{aligned}$$

As the number of ensemble trees $B \to \infty$, the second term $\frac{1-\rho}{B}\sigma^2 \to 0$, bounding variance strictly by $\rho\sigma^2$. Random feature subspace selection reduces ρ, achieving superior variance suppression.